

Reproducible methodology for stress-testing astronomical photometric workflows under compound contamination

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October 3, 2026

Abstract

AMPV is a reproducible validation methodology for mapping the reliability of astronomical photometric workflows under controlled compound contamination. The primary comet-like experiment evaluates workflows A–E on five reference-contrast levels and three contamination levels, using 300 matched stochastic identities to form 4,500 condition-specific scenes and 22,500 workflow evaluations. A pre-specified extension, internally frozen before outcome generation, evaluates a materially different compact-PSF task family with 300 identities and 18,000 evaluations of blind workflows A, B, C, and E. Performance is summarized with signed and absolute relative error and a predefined vulnerability indicator at $|e| > 0.5$, with sample-identity bootstrap uncertainty and threshold sensitivity. In the comet-like family, all 60 adjacent-reference-contrast paired comparisons showed lower absolute error at the next tested contrast level, whereas contamination effects were not uniform across workflow–contrast cells. In contaminated compact-PSF cells, all 32 adjacent-reference-contrast comparisons showed the same improving direction, while all 40 tested contamination contrasts were adverse. Workflow-relative behavior and vulnerability boundaries differed between the two families; the compact-PSF advantage of Gaussian model fitting is interpreted as a task-specific model-match effect rather than universal superiority.

28 The two controlled synthetic families therefore support AMPV as a
29 method for recovering task-conditional, cell-resolved reliability struc-
30 ture rather than assigning a global workflow ranking. The study re-
31 mains single-fidelity and synthetic, so the reported boundaries are
32 restricted to the tested validation domains and do not constitute ob-
33 servational validation.

34 **Keywords:** astronomical photometry, validation, synthetic perturbations,
35 robustness, reliability mapping, reproducibility

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37 1 Introduction

38 1.1 Motivation

39 A single aggregate performance summary cannot by itself show where a work-
40 flow remains reliable or where it fails across controlled changes in observa-
41 tional difficulty.

42 Astronomical photometry and source-extraction workflows have been eval-
43 uated with controlled simulations, injected sources, and survey-level valida-
44 tion. Comparisons on simulated images can expose condition-dependent be-
45 havior, including degradation in crowded or overdense scenes [1]. Injection
46 frameworks such as HSC SynPipe place synthetic sources into real survey
47 images and have revealed blending-dependent photometric caveats [2]. Foun-
48 dational tools such as SExtractor provide automated source detection, de-

49 blending, measurement, and classification capabilities [3], while large survey
 50 releases such as Gaia EDR3 include dedicated photometric validation and
 51 systematic-quality assessment [4]. End-to-end simulators such as PhoSim
 52 provide another route for studying pipeline behavior and systematics, while
 53 still requiring comparison with real data to establish fidelity in observational
 54 use [5]. The AMPV study is designed around that gap in evaluation struc-
 55 ture: instead of treating validation as a single ranking problem, it maps
 56 workflow behavior across a frozen stress grid and retains the cell-level evi-
 57 dence needed to identify reliability boundaries.

58 Astronomical image simulation and injection-based validation also pro-
 59 vide mature precedents for testing image-analysis systems. GalSim provides
 60 a general-purpose, high-precision framework for rendering synthetic astro-
 61 nomical sources [8]. Survey pipelines have been characterized by insert-
 62 ing synthetic sources into observational images: HSC SynPipe propagates
 63 injected objects through the HSC processing chain [2], while Balrog uses
 64 large-scale injections into DES single-epoch images to sample selection and
 65 photometric biases and the survey transfer function [9]. For task-specific al-
 66 gorithm evaluation, the Blending ToolKit combines simulated blended-galaxy
 67 images, standardized algorithm interfaces, and common metrics to support

68 principled comparisons of detection and deblending methods [10]. AMPV is
69 positioned alongside, rather than prior to, these approaches. Its emphasis
70 in the present study is a matched compound-contamination stress grid with
71 preserved stochastic identities, explicit separation of blind and privileged-
72 information workflow classes, and cell-resolved reliability and operational-
73 vulnerability surfaces used to report boundaries only on the tested grid.

74 The present experiment uses a controlled synthetic environment so that
75 ground truth, perturbation strength, and repeated scene realization can be
76 fixed explicitly. [5] This makes it possible to separate changes in workflow
77 output from changes in the underlying stochastic realization and to preserve
78 a reproducible audit trail from scene generation through numerical summary.

79 **1.2 From aggregate comparison to reliability mapping**

80 AMPV is framed here as a validation methodology rather than as an ad-
81 ditional photometric estimator. The objects being evaluated are workflows
82 A–E; AMPV supplies the experimental design, matched-scene structure, nu-
83 merical ledger, and boundary-oriented interpretation used to assess them.

84 The frozen V2 design crosses five reference-contrast levels with three con-
85 tamination levels and evaluates every workflow on the same matched scenes.

86 This structure supports two complementary views of performance: continu-
87 ous error behavior and a threshold-based vulnerability indicator. Together,
88 those views describe where error remains small, where vulnerability becomes
89 nonzero or increases, and how those patterns depend on the tested stress
90 coordinates.

91 **1.3 Relationship to CAPS**

92 AMPV and CAPS serve different roles in the present work. AMPV is the
93 general validation methodology used to organize stress testing, reliability
94 mapping, and boundary analysis. CAPS is a controlled synthetic implemen-
95 tation environment that can support such experiments. The V2 production
96 used in this manuscript was independently realized, audited, and canonically
97 frozen within the AMPV V2 project tree; the earlier CAPS V1 production
98 outputs are not used as the canonical evidence for the present results.

99 This distinction is important because the scientific contribution of the
100 present study is not a renamed workflow or a simple continuation of an
101 earlier ranking exercise. The emphasis is the reproducible methodology for
102 characterizing reliability across controlled perturbation space.

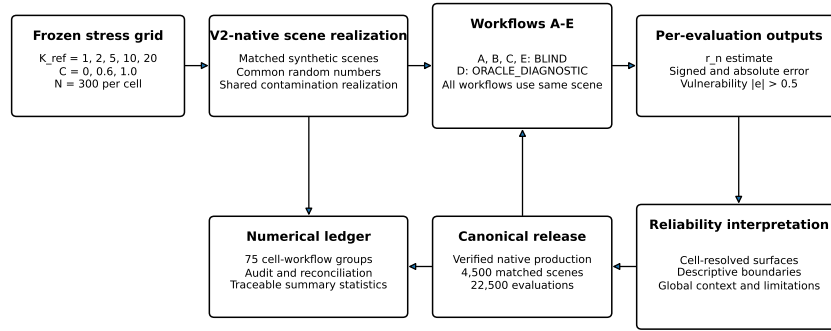
1.4 Study objectives

The present study has four objectives:

1. to define and execute a frozen matched-scene stress grid over reference contrast and contamination level for workflows A–E in the primary comet-like family, with a separately frozen blind-workflow A/B/C/E extension in the compact-PSF family;
2. to quantify both continuous error behavior and threshold-defined vulnerability at the cell level rather than relying only on pooled summaries;
3. to identify descriptive reliability boundaries while preserving workflow information-class distinctions, including the diagnostic role of privileged-information workflow D; and
4. to establish a fully traceable production-to-ledger-to-canonical workflow so that every manuscript number can be linked back to frozen V2-native evidence.

The present study is single-fidelity but now contains two separately versioned controlled synthetic task families. The canonical `COMET_LIKE_PRIMARY` experiment is retained unchanged, and a pre-registered `COMPACT_PSF_PRIMARY` extension tests whether the AMPV reliability-mapping structure remains in-

121 formative after a material change in source morphology and estimand. Multi-
 122 fidelity dependence remains outside the evidence supported by the current
 123 study.



AMPV = validation methodology; CAPS = one controlled synthetic implementation environment.
 Current canonical experiment: single fidelity (F0). Reliability statements are restricted to the frozen tested grid.

Figure 1: AMPV validation framework.

124 1.5 Scope of external context

125 External context is limited to source-level-audited literature. Prior-art state-
 126 ments are restricted to controlled simulation and source injection, photo-
 127 metric validation, source-extraction comparison, common-random-number
 128 methodology, and validation-domain framing supported by Refs. [1, 2, 3, 4,
 129 5, 6, 7, 8, 9, 10]. Claims about the present AMPV results remain grounded
 130 in the frozen V2 evidence chain.

131 2 Methods

132 2.1 AMPV validation design and statistical unit

133 AMPV is used here to map workflow reliability under controlled compound
134 contamination. The experiment contains 300 base stochastic sample identi-
135 ties. Each identity is propagated through 15 condition cells formed by five
136 reference-contrast levels, $\kappa_{\text{ref}} = 1, 2, 5, 10, 20$, crossed with three contami-
137 nation levels, $C = 0, 0.6, 1.0$. This produces 4,500 condition-specific scene
138 instances. Each scene instance is evaluated by workflows A–E, giving 22,500
139 workflow evaluations and 75 cell–workflow groups with 300 evaluations per
140 group. The task family is `COMET_LIKE_PRIMARY` at fidelity level F0.

141 The same sample identity preserves r_n , interference seeds, and the un-
142 scaled structured-contamination realization across all 15 cells. Thus the
143 4,500 condition-specific instances are not treated as 4,500 statistically in-
144 dependent base scenes. This common-random-number design permits paired
145 comparisons across stress conditions and workflows while reducing variation
146 due solely to unrelated stochastic draws.

147 **2.2 Synthetic signal and compound contamination**

148 The implemented comet-like source is generated on a 256×256 pixel image.
149 The central nucleus is represented by a normalized Gaussian component with
150 $\sigma = 1.5$ pixels and is combined with a radial coma component proportional to
151 r_n^2/ρ^2 after the implemented central capping and normalization. The canon-
152 ical benchmark uses a nucleus fraction of 0.3 and a coma fraction of 0.7. This
153 construction is an intentionally controlled synthetic task and is not presented
154 as a complete physical radiative-transfer model.

155 Compound interference contains six implemented families: I1, sky gra-
156 dient/spatial mottling/background galaxies; I2, a dust jet and anti-solar
157 tail; I3, sparse cosmic-ray-like events; I4, field-star Gaussian PSFs; I5, low-
158 frequency detector baseline drift; and I6, Gaussian readout noise. The struc-
159 tured I1–I5 realization is shared for a given sample identity and scaled by C ,
160 whereas I6 is retained separately. The realized r_n values in the production
161 span 0.310041–4.988671.

162 **2.3 Reference-contrast calibration**

163 Reference contrast is controlled through a source-gain calibration. For the
164 reference regime, $r_n = 1$ km and the clean source flux is measured in a

165 30-pixel aperture. The reference noise denominator is the I6 readout-noise
 166 standard deviation multiplied by the square root of the aperture pixel count.
 167 For each target κ_{ref} , source gain is selected so that the reference clean-
 168 aperture flux attains the requested contrast. The resulting gains for $\kappa_{\text{ref}} =$
 169 1, 2, 5, 10, 20 are 13.9532646896, 27.9065293792, 69.766323448, 139.532646896,
 170 and 279.065293792, respectively. The same gain is used across the three con-
 171 tamination levels at a given target contrast. The production also records the
 172 realized sample-specific contrast separately from the configured κ_{ref} .

173 **2.4 Photometric workflows and information constraints**

174 Workflow A is a blind 30-pixel-radius aperture sum with no background sub-
 175 traction. Workflow B is blind aperture photometry using a 30-pixel circular
 176 aperture and a fixed 50–90 pixel annulus; the annular mean is used for back-
 177 ground subtraction. Workflow C is blind aperture photometry with the same
 178 aperture and background ring but uses a deterministic RANSAC-style back-
 179 ground estimator (seed 20260814, 200 iterations, sample size 30, and an inlier
 180 threshold of 0.2

181 , σ of the background pixels).

182 Workflow D is an `ORACLE_DIAGNOSTIC`, not an information-equivalent

183 blind estimator. It uses the true r_n to select an adaptive background an-
184 nulus with inner radius $\max[\lceil 11r_n \rceil, 35]$ pixels, subject to image-geometry
185 limits, a 40-pixel annulus width, and median background estimation. It is
186 therefore interpreted as a privileged-information diagnostic reference.

187 Workflow E is blind model fitting with a circular two-dimensional Gaus-
188 sian plus constant background on a local image patch. The least-squares
189 parameters are source center, amplitude, Gaussian width, and background;
190 an analytic Jacobian is used, the Gaussian width is bounded to 1–20 pixels,
191 and the optimization limit is 120 function evaluations. Fitted flux is $2\pi|A|\sigma^2$.

192 Each workflow has a separate noiseless calibration curve based on 40
193 log-spaced r_n values from 0.1 to 10 km. Measured flux is mapped back to
194 r_n by log-flux interpolation. Calibration is constructed separately for each
195 workflow and source-gain level.

196 **2.5 Pre-registered compact-PSF extension**

197 A second controlled family, `COMPACT_PSF_PRIMARY`, was pre-registered and
198 executed as a separately versioned extension after the comet-like produc-
199 tion had been frozen. It contains 300 base identities crossed with the same
200 five reference-contrast targets and three contamination levels. Because no

201 scientifically meaningful privileged-information analogue of workflow D was
202 identified before production, the extension evaluates blind workflows A, B,
203 C, and E, giving 18,000 workflow evaluations.

204 The source is a centered circular Gaussian PSF with independently sam-
205 pled width $\sigma_{\text{PSF}} = 1.5\text{--}4.0$ pixels and integrated truth flux F_{true} . A two-
206 dimensional Latin-hypercube design (seed 20261003) samples PSF width
207 and a narrow intrinsic-amplitude factor of 0.8–1.2. The comet-specific dust-
208 jet/tail interference I2 is not reused. It is replaced, before outcome gen-
209 eration, by I2P: a task-neutral asymmetric elliptical local contaminant with
210 geometry and amplitude fixed relative to a reference signal scale and indepen-
211 dent of sample F_{true} and σ_{PSF} . The remaining stress construction preserves
212 matched sample identities across conditions.

213 For this family, workflows A–C retain the same aperture/background def-
214 initions, whereas workflow E fits the circular Gaussian plus constant back-
215 ground. The estimand is integrated flux, so relative error is $(F^{\text{est}} - F_{\text{true}})/F_{\text{true}}$.
216 A clean-domain audit showed only numerical-precision-level error for A–C
217 and E, so no post-hoc calibration surface was introduced. The produc-
218 tion and statistical analysis were frozen before scientific interpretation; the
219 latter used 2,000 sample-identity bootstrap replicates with seed 20261004.

220 The same primary $\tau = 0.5$ vulnerability definition and the same threshold-
 221 sensitivity grid were retained.

222 **2.6 Error quantities and operational vulnerability cri-** 223 **terion**

224 Signed relative error is defined with the family-specific estimand. For COMET_LIKE_PRIMARY,

$$e = (r_n^{\text{est}} - r_n^{\text{true}})/r_n^{\text{true}},$$

225 whereas for COMPACT_PSF_PRIMARY,

$$e = (F^{\text{est}} - F_{\text{true}})/F_{\text{true}}.$$

226 In both families, absolute error is $|e|$. Continuous error is the primary mea-
 227 sure of estimation behavior. The predefined vulnerability indicator uses
 228 $|e| > \tau$ with the primary frozen operational threshold $\tau = 0.5$, corresponding
 229 to a 50% relative-error cutoff. This threshold is not treated as a universal
 230 astronomical standard; its robustness is examined by sensitivity analyses at
 231 $\tau = 0.25, 0.4, 0.5, 0.6, 0.75$, and 1.0.

2.7 Paired uncertainty and robustness analyses

Uncertainty analyses use the 300 sample identities within each task family as the resampling units. A cluster-bootstrap replicate samples these identities with replacement and carries each selected identity’s complete cross-condition and cross-workflow block within that family, preserving the matched-design dependence. The comet-like analysis used 2,000 replicates with fixed seed 20261002; the separately frozen compact-PSF analysis used 2,000 replicates with fixed seed 20261004. These analyses provide 95% intervals for cell-level median absolute error, median signed error, vulnerability, and paired contrasts. In the comet-like family, blind-workflow contrasts are evaluated within matched $(\kappa_{\text{ref}}, C, \text{sampleid})$ instances and workflow D is kept separate because of its oracle information class. The compact-PSF extension contains only blind workflows A, B, C, and E.

Workflow E has complete optimization diagnostic records for all 4,500 evaluations. The optimizer reports **success=True** for 4,497 evaluations and **success=False** with status 0 for three. The production validity rule classified computational failure on exception rather than optimizer status alone. A sensitivity analysis therefore recomputed affected summaries after excluding these three cases without altering their canonical validity labels.

251 **2.8 Reproducibility and provenance**

252 For each task family, the experimental design, seed namespace, workflow
253 definitions, and stress grid were frozen before that family’s production. The
254 original comet-like canonical release was not modified when the separately
255 versioned compact-PSF extension was added. Numerical summaries and fig-
256 ures were generated from the corresponding verified production rows and
257 audited derived-analysis records without altering either production output.
258 Detailed hashes, audit gates, and engineering provenance are retained in the
259 reproducibility archive rather than repeated in the scientific narrative.

260 **2.9 Ethical applicability and AI-assisted workflow**

261 The present work is a computational validation study based on synthetic
262 astronomical scenes and simulation outputs. The AMPV V2 evidence chain
263 contains no human-participant, animal-subject, clinical-trial, or personally
264 identifiable data. On that project-evidence basis, institutional human/animal
265 research approval and informed consent are not applicable to the current
266 study scope. **[AUTHOR CONFIRMATION REQUIRED BEFORE**
267 **SUBMISSION]**

268 ChatGPT (OpenAI) was used as an assistive tool during manuscript

269 preparation and reproducibility-workflow development. Its use included lan-
270 guage editing; drafting and reviewing Python, PowerShell, and LaTeX work-
271 flow scripts; organizing provenance and audit records; supporting literature-
272 audit and citation-workflow organization; and assisting with figure-layout
273 and manuscript-assembly tasks. AI assistance did not replace the AMPV V2
274 numerical evidence chain. The authors remain responsible for independent
275 verification of scientific content, source use, code execution records, numeri-
276 cal results, interpretations, and the final manuscript text. ChatGPT is not
277 an author.

278 **3 Results**

279 The primary comet-like analysis contains 300 base stochastic sample identi-
280 ties propagated through 15 condition cells, yielding 4,500 condition-specific
281 scene instances and 22,500 workflow evaluations organized into 75 cell-workflow
282 groups. Its grid comprises five reference-contrast levels (reference contrast =
283 1, 2, 5, 10, 20), three contamination levels ($C = 0, 0.6, 1.0$), and work-
284 flows A–E. The comet-like values reported in the following subsections are
285 derived from the verified canonical release and audited numerical ledger; the

compact-PSF extension is reported separately from its independently frozen production.

3.1 Global descriptive behavior across the frozen grid

Across the full frozen grid, the workflow-level summaries were:

- Workflow A: $N=4500$, median absolute error=0.019, IQR=0.0051–0.073, median signed error=0.012, vulnerability rate=5.5%.
- Workflow B: $N=4500$, median absolute error=0.015, IQR=0.0040–0.060, median signed error=0.0057, vulnerability rate=4.8%.
- Workflow C: $N=4500$, median absolute error=0.061, IQR=0.016–0.232, median signed error=8.852e-04, vulnerability rate=15.3%.
- Workflow D: $N=4500$, median absolute error=0.015, IQR=0.0038–0.058, median signed error=0.0025, vulnerability rate=5.0%.
- Workflow E: $N=4500$, median absolute error=0.0034, IQR=8.988e-04–0.013, median signed error=8.213e-04, vulnerability rate=1.8%.

These aggregate summaries pool all 15 experimental cells and are used only as a compact descriptive overview. Cell-resolved behavior is the primary basis for discussing reliability boundaries. The pooled workflow-level summaries are shown in Fig. 2.

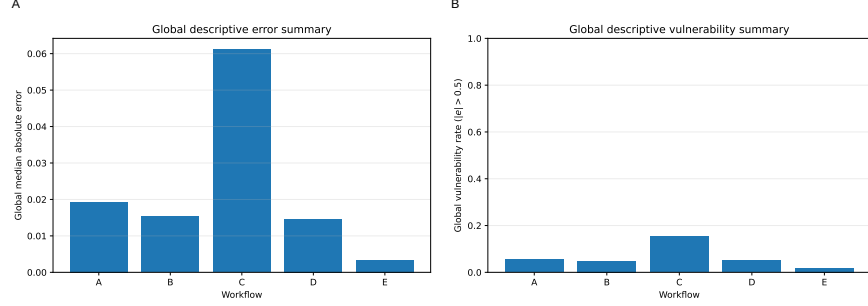


Figure 2: Global pooled descriptive context for the primary comet-like experiment.

3.2 Cell-resolved error and vulnerability landscape

The 15 experimental cells show substantial variation in median absolute error and in the fraction of evaluations exceeding the predefined vulnerability threshold ($|e| > 0.5$). The cell-resolved median absolute-error landscape is shown in Fig. 3, and the corresponding vulnerability landscape is shown in Fig. 4. For each cell, the observed numerical span across workflows is:

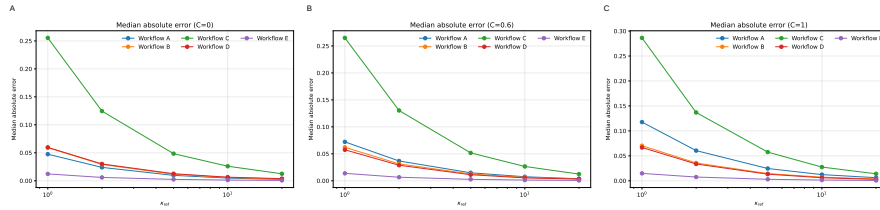


Figure 3: Cell-resolved median absolute-error landscape for the primary comet-like experiment.

- Reference contrast=1, C=0: median $|e|$ spans 0.012–0.255 (lower end-

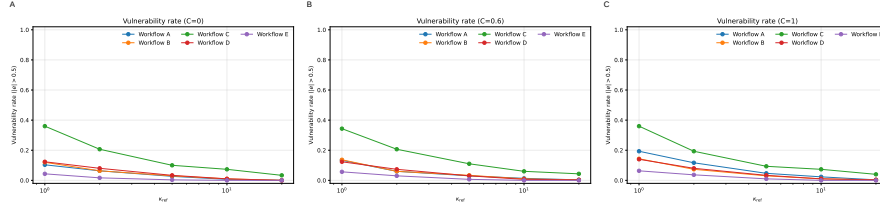


Figure 4: Cell-resolved vulnerability landscape for the primary comet-like experiment at the primary threshold.

- point: E; upper endpoint: C); vulnerability spans 4.3%–36.0% (lower
 endpoint: E; upper endpoint: C).
- Reference contrast=2, C=0: median $|e|$ spans 0.0060–0.125 (lower end-
 point: E; upper endpoint: C); vulnerability spans 1.7%–20.7% (lower
 endpoint: E; upper endpoint: C).
 - Reference contrast=5, C=0: median $|e|$ spans 0.0024–0.049 (lower end-
 point: E; upper endpoint: C); vulnerability spans 0.3%–10.0% (lower
 endpoint: E; upper endpoint: C).
 - Reference contrast=10, C=0: median $|e|$ spans 0.0012–0.026 (lower
 endpoint: E; upper endpoint: C); vulnerability spans 0.0%–7.3% (lower
 endpoint: E; upper endpoint: C).
 - Reference contrast=20, C=0: median $|e|$ spans 6.177e-04–0.012 (lower
 endpoint: E; upper endpoint: C); vulnerability spans 0.0%–3.3% (lower
 endpoint: A,B,D,E; upper endpoint: C).

- 325 • Reference contrast=1, C=0.6: median $|e|$ spans 0.014–0.265 (lower end-
326 point: E; upper endpoint: C); vulnerability spans 5.7%–34.3% (lower
327 endpoint: E; upper endpoint: C).
- 328 • Reference contrast=2, C=0.6: median $|e|$ spans 0.0066–0.130 (lower
329 endpoint: E; upper endpoint: C); vulnerability spans 3.0%–20.7%
330 (lower endpoint: E; upper endpoint: C).
- 331 • Reference contrast=5, C=0.6: median $|e|$ spans 0.0027–0.052 (lower
332 endpoint: E; upper endpoint: C); vulnerability spans 0.7%–11.0%
333 (lower endpoint: E; upper endpoint: C).
- 334 • Reference contrast=10, C=0.6: median $|e|$ spans 0.0013–0.027 (lower
335 endpoint: E; upper endpoint: C); vulnerability spans 0.0%–6.0% (lower
336 endpoint: E; upper endpoint: C).
- 337 • Reference contrast=20, C=0.6: median $|e|$ spans 6.540e-04–0.012 (lower
338 endpoint: E; upper endpoint: C); vulnerability spans 0.0%–4.3% (lower
339 endpoint: E; upper endpoint: C).
- 340 • Reference contrast=1, C=1: median $|e|$ spans 0.015–0.286 (lower end-
341 point: E; upper endpoint: C); vulnerability spans 6.3%–36.0% (lower
342 endpoint: E; upper endpoint: C).
- 343 • Reference contrast=2, C=1: median $|e|$ spans 0.0075–0.137 (lower end-

point: E; upper endpoint: C); vulnerability spans 3.7%–19.3% (lower
 endpoint: E; upper endpoint: C).

- Reference contrast=5, C=1: median $|e|$ spans 0.0030–0.057 (lower end-
 point: E; upper endpoint: C); vulnerability spans 1.0%–9.3% (lower
 endpoint: E; upper endpoint: C).
- Reference contrast=10, C=1: median $|e|$ spans 0.0015–0.027 (lower
 endpoint: E; upper endpoint: C); vulnerability spans 0.0%–7.3% (lower
 endpoint: E; upper endpoint: C).
- Reference contrast=20, C=1: median $|e|$ spans 7.387e-04–0.014 (lower
 endpoint: E; upper endpoint: C); vulnerability spans 0.0%–4.0% (lower
 endpoint: E; upper endpoint: C).

The corresponding cell-resolved values are listed below.

Complete cell-level values for all 75 cell–workflow groups, including me-
 dian absolute error, vulnerability rate, median signed error, and realized
 contrast, are provided in Table S1.

3.3 Signed-error behavior

Median signed error is retained separately from absolute error to distinguish
 error magnitude from direction. Figure S1 shows the frozen signed-error

362 curves across the three contamination levels. Complete cell-level signed-
363 error values are included in Table S1. Directional structure is interpreted
364 only together with error magnitude, workflow information class, and the
365 corresponding stress-grid cell.

366 **3.4 Reliability-boundary framing**

367 The frozen grid is reported as a reliability landscape rather than as a single
368 global ranking. The cell-resolved summaries show that error magnitude and
369 vulnerability vary with both reference contrast and contamination level **C**,
370 and that the magnitude of separation among workflows is not constant across
371 the grid.

372 At this stage, the data support reporting: - where median absolute errors
373 remain small or become large; - where vulnerability rates become nonzero
374 or increase; - how those patterns shift across reference contrast and **C**; - how
375 signed-error behavior differs from absolute-error behavior.

376 Interpretive explanations for why those patterns occur are reserved for
377 the Discussion.

3.5 Paired uncertainty and robustness

Cluster-bootstrap uncertainty was evaluated with the 300 base sample identities as the resampling units, preserving each identity’s complete cross-condition and cross-workflow block. The resulting intervals are reported with the derived-analysis tables. Paired blind-workflow contrasts confirm that the separations visible in the cell-resolved landscape are condition dependent rather than reducible to a single pooled ordering. In particular, the paired absolute-error difference between workflows C and E is positive in all 15 tested cells, with the 95% cluster-bootstrap interval above zero in every cell. The corresponding A–E and B–E differences are also positive in all 15 cells, whereas A–B changes with contamination condition and some low-contrast intervals include zero.

Paired stress-axis contrasts sharpen this condition-dependent interpretation for the comet-like family. For contamination, 35 of 50 higher- C versus $C = 0$ contrasts had 95% sample-identity bootstrap intervals entirely above zero, 13 included zero, and two were below zero; both negative intervals occurred for privileged-information workflow D at $\kappa_{\text{ref}} = 20$. Thus contamination predominantly increased absolute error but not uniformly across all workflow–contrast cells. In contrast, all 60 adjacent- κ paired contrasts had

397 intervals entirely below zero for $\Delta|e| = |e|_{\kappa_{\text{high}}} - |e|_{\kappa_{\text{low}}}$, indicating lower
 398 absolute error at each successive tested reference-contrast level across all
 399 workflows and contamination settings.

400 Threshold sensitivity confirms that the numerical location of a threshold-
 401 defined boundary depends on the operational cutoff. The primary $\tau = 0.5$
 402 analysis exactly reproduces the predefined boundary table below. At $\tau =$
 403 0.4, the A boundary at $C = 1$ moves from $\kappa_{\text{ref}} = 5$ to 10; at $\tau = 0.6$,
 404 the E boundary at $C = 0.6$ moves from 2 to 1 while the A–D primary
 405 boundaries are unchanged. More stringent or permissive cutoffs produce
 406 larger shifts. Continuous error surfaces are therefore treated as primary
 407 evidence and threshold boundaries as an operational summary.

408 Excluding the three Workflow-E evaluations with optimizer `success=False`
 409 changes the affected cell median absolute errors by at most 9.02×10^{-5} , in-
 410 dicating no material effect on the reported reliability landscape.

411 **3.6 Pre-registered compact-PSF extension**

412 The second family contains 300 base identities and 18,000 workflow eval-
 413 uations across 60 condition–workflow groups. All 40 paired contamination
 414 contrasts comparing $C = 0.6$ or 1 with $C = 0$ had 95% bootstrap inter-

415 vals above zero. Of 48 adjacent- κ contrasts, 32 had intervals below zero;
416 the remaining 16 were all $C = 0$ clean cells, where error was already at the
417 numerical floor. Restricting the comparison to contaminated cells therefore
418 gives 32 of 32 adjacent- κ intervals below zero.

419 Workflow-relative structure also differed from the comet-like family. Within
420 contaminated compact-PSF cells, workflow E had lower paired absolute er-
421 ror than A, B, and C throughout the tested grid; A had lower paired error
422 than B and C, whereas B-C intervals included zero throughout those cells.
423 The E result is interpreted as a family-specific model-match effect because
424 the source truth is itself a circular Gaussian, not as evidence of universal
425 workflow superiority. Threshold sensitivity placed operational vulnerability
426 mainly in the low-contrast, high-contamination region; by $\kappa_{\text{ref}} \geq 5$, vulner-
427 ability was essentially absent even at $\tau = 0.25$, apart from one 1/300 event
428 for workflow C at $\kappa_{\text{ref}} = 5, C = 1$.

429 4 Discussion

430 4.1 Reliability is a landscape rather than a single ag- 431 gregate score

432 The frozen AMPV V2 results are most naturally interpreted as a reliability
433 landscape over reference contrast and contamination level, rather than as a
434 single global ordering of workflows. The global summaries are:

- 435 • Workflow A: global median $|e|=0.019$; global vulnerability=5.5%; global
436 median signed $e=0.012$.
- 437 • Workflow B: global median $|e|=0.015$; global vulnerability=4.8%; global
438 median signed $e=0.0057$.
- 439 • Workflow C: global median $|e|=0.061$; global vulnerability=15.3%; global
440 median signed $e=8.852e-04$.
- 441 • Workflow D: global median $|e|=0.015$; global vulnerability=5.0%; global
442 median signed $e=0.0025$.
- 443 • Workflow E: global median $|e|=0.0034$; global vulnerability=1.8%; global
444 median signed $e=8.213e-04$.

445 These aggregate values are useful for orientation, but they pool cells whose

446 error and vulnerability behavior can differ materially. For that reason, the
447 cell-resolved surfaces remain the primary evidence for identifying stable and
448 vulnerable operating regions.

449 Across the frozen reference-contrast grid, median absolute error and vul-
450 nerability generally decline as reference contrast increases for all workflows
451 and contamination levels. The strongest residual vulnerability occurs for
452 workflow C at low contrast, while workflow E reaches low vulnerability at
453 lower tested contrast levels. The complete workflow-by-contamination mono-
454 tonic trend listing is provided in Supplementary Note S1.

455 This pattern supports the methodological use of AMPV as a boundary-
456 mapping framework: the relevant question is not only whether two workflows
457 differ on average, but also whether their error behavior remains stable as the
458 controlled stressors move across the grid.

459 **4.2 Contamination changes the shape of the reliability** 460 **surface**

461 At fixed reference contrast, contamination modifies the observed error and
462 vulnerability patterns in a workflow-dependent manner. The effect is strongest
463 in the low-contrast regime and is not uniformly monotonic for every workflow–

contrast combination. Consequently, contamination cannot be summarized independently of reference contrast or information class. The complete cell-by-cell contamination trend listing is provided in Supplementary Note S2.

4.3 A threshold-based view exposes operational boundaries

Using the predefined vulnerability threshold $|e| > 0.5$, the lowest tested reference contrast at which vulnerability is $\leq 5\%$ is summarized below.

Workflow	C=0	C=0.6	C=1
A	5	5	5
B	5	5	5
C	20	20	20
D	5	5	5
E	1	2	2

This boundary table should be interpreted only within the tested grid. [6] It does not imply behavior between untested reference-contrast levels, and it should not be extrapolated beyond the frozen contamination range.

4.4 Information class matters for interpretation

The frozen production metadata distinguish workflow D as the oracle-diagnostic information class, whereas the remaining workflows are blind. Therefore, numerical differences involving D should not be interpreted as if all workflows operate under the same information constraints. Workflow D is used as a diagnostic reference for what becomes possible with privileged information rather than as a directly interchangeable blind estimator.

This distinction is methodological rather than cosmetic: reliability claims should remain conditional on the information available to a workflow.

4.5 Signed error and absolute error answer different questions

The analysis retains both absolute and signed error. Absolute error describes magnitude and is therefore used for vulnerability mapping, whereas signed error preserves direction. The signed-error figure is retained as supplementary material so that directional structure remains inspectable without replacing the primary magnitude-based reliability view.

A directional pattern should not be described as improved or degraded

491 performance solely from its sign. The interpretation must remain tied to
492 error magnitude, information class, and the relevant cell of the stress grid.

493 **4.6 Cross-family structure is reproducible but task-** 494 **dependent**

495 The extension does not reproduce a fixed workflow ordering, nor was that the
496 pre-registered transfer criterion. Across both controlled families, increasing
497 tested reference contrast is associated with lower paired absolute error wher-
498 ever nontrivial error remains, compound contamination changes the reliabil-
499 ity landscape, and matched identities resolve workflow-dependent structure.
500 However, the detailed contamination response, workflow ordering, and vul-
501 nerability boundaries differ between the comet-like and compact-PSF tasks.

502 This heterogeneity is methodologically informative. In the comet-like
503 family, contamination is not uniformly monotonic and A–B behavior changes
504 with contamination. In the compact-PSF family, all tested contamination
505 contrasts are adverse and the Gaussian-fitting workflow benefits from direct
506 model match. AMPV should therefore be interpreted as a framework for
507 estimating task-conditional reliability structure, not for assigning a task-
508 independent score or universal workflow ranking.

509 4.7 What the present experiment does and does not 510 establish

511 The study contains two frozen single-fidelity controlled experiments. The
512 original canonical comet-like production establishes the tested reliability land-
513 scape for `COMET_LIKE_PRIMARY`, the five reference-contrast levels, the three
514 contamination levels, and workflows A–E under its V2-native matched stochas-
515 tic design. The separately versioned compact-PSF extension establishes the
516 corresponding tested landscape for `COMPACT_PSF_PRIMARY` with blind work-
517 flows A, B, C, and E. Neither production is retroactively redefined by the
518 other.

519 It does **not** by itself establish fidelity dependence across multiple simu-
520 lation fidelities. [5] Any manuscript claim about fidelity dependence would
521 require additional, explicitly authorized multi-fidelity experiments or must
522 be framed as future work rather than as a result of the present production.

523 Likewise, the synthetic environment provides controlled access to ground
524 truth and reproducible perturbations, but behavior on observational datasets
525 outside the tested validation domain still requires external validation [5, 6].
526 The value of the framework is therefore methodological: it provides a repro-

527 ducible way to expose where a workflow remains stable, where it becomes
528 vulnerable, and which controlled stressors are associated with those changes.

529 **4.8 Relationship between AMPV and CAPS**

530 AMPV and CAPS are kept conceptually distinct. AMPV is the general
531 validation methodology developed around reliability mapping and boundary
532 analysis, whereas CAPS is one controlled synthetic implementation environ-
533 ment that can support such validation experiments. The present V2 pro-
534 duction is independently realized and canonically frozen within the AMPV
535 V2 project tree; CAPS V1 production outputs are not used as the present
536 canonical evidence.

537 This framing avoids treating AMPV as merely a renamed workflow or a
538 simple continuation of the earlier ranking study.

539 **4.9 Limitations and next validation steps**

540 The present results are bounded by the tested grids, the two controlled syn-
541 thetic task families, the current fidelity level, and their synthetic forward-
542 model assumptions. The current analysis also uses a predefined vulnera-
543 bility threshold of 0.5; alternative thresholds may change the location of

544 threshold-defined boundaries even when the underlying continuous error sur-
545 face is unchanged. The explicit sensitivity analysis confirms this dependence
546 and therefore supports interpreting the threshold table as an operational
547 diagnostic rather than a physical transition.

548 Useful next validation steps include testing additional fidelity levels, ex-
549 panding the stress grid where transitions appear sharp, and evaluating the
550 framework with observationally anchored injections, real-image test sets, or
551 independently implemented community pipelines. [6] [5] These extensions
552 should be run as new, versioned experiments rather than retrofitted into the
553 frozen canonical release.

554 Literature-dependent context is limited to the source-level-audited ref-
555 erences. Interpretation of the present numerical results remains separated
556 from prior-art context and is restricted to the frozen tested grids, informa-
557 tion classes, and single-fidelity production records.

558 **5 Conclusions**

559 The primary frozen AMPV V2 comet-like experiment demonstrates a re-
560 producible way to evaluate photometric workflows as a reliability landscape

rather than as a single aggregate ranking. Across 300 base stochastic identities, 4,500 condition-specific scene instances, and 22,500 workflow evaluations, that primary analysis retains cell-resolved error magnitude, signed error, vulnerability, workflow information class, and matched-scene provenance across the tested reference-contrast and contamination grid.

The results support boundary-oriented reporting: workflow behavior varies across the frozen stress grid, and pooled summaries alone do not preserve the full structure of those changes. The common-random-number design follows an established simulation-comparison strategy [7], while reproducibility and traceability in the present study are provided by the audited canonical release chain.

The pre-specified, internally frozen extension broadens the evidence from one to two materially distinct controlled synthetic photometric task families. Both recover cell-resolved stress-dependent reliability structure, while their workflow-relative behavior and operational boundaries differ. This supports task-conditional reliability mapping rather than a universal ranking interpretation.

The present evidence is intentionally bounded. It is a single-fidelity synthetic study and therefore does not establish multi-fidelity dependence. More

generally, the synthetic evidence should not be assumed to generalize un-
changed to observational datasets outside the tested validation domain [5, 6].
Those questions require separately versioned validation experiments rather
than reinterpretation of the frozen release.

Within that scope, AMPV provides a methodology for moving from av-
erage workflow comparison toward explicit mapping of stable and vulnerable
operating regions under controlled perturbation.

Disclosures

The author declares no competing interests.

Code, Data, and Materials Availability

EXTERNAL ACTION REQUIRED. A permanent AMPV V2 reposi-
tory record, public record URL, and DOI or other persistent identifier have
not yet been assigned. The local reproducibility archive candidate is not
a substitute for the final public Data/Code availability statement. Materi-
als availability is expected to be not applicable to this computational study,
subject to author confirmation.

596 Acknowledgments

597 This research received no external funding.

598 During the preparation of this work, the author used OpenAI Chat-
599 GPT for English-language editing, manuscript organization, and reference-
600 formatting tasks. It also provided technical assistance for parts of Python
601 code development, review, testing, and reproducibility-audit workflows within
602 the AMPV study. All scientific results were derived from frozen, version-
603 controlled computational outputs and were not AI-generated. Key numeri-
604 cal values were independently verified through reproduction runs and reverse
605 numerical-provenance audits. Figures, tables, statistical outputs, references,
606 and manuscript conclusions were cross-checked against the corresponding
607 frozen records. All decisions regarding study design, methodological choices,
608 scientific interpretation, and manuscript submission were made by the au-
609 thor. The author reviewed the final code and manuscript and takes full
610 responsibility for the accuracy, integrity, and content of the work.

611 **Author Contributions**

612 **AUTHOR INPUT REQUIRED.** Assign CRediT contributor roles to
613 every listed author before submission.

614 **Originality and Submission Status**

615 **AUTHOR CONFIRMATION REQUIRED.** Confirm that AMPV V2
616 is original work and is not under concurrent journal consideration. Confirm
617 separately whether AMPV V2 itself has previously been submitted to or
618 declined by an SPIE journal.

619 CAPS V1 is related prior work, but AMPV V2 is a distinct validation-
620 methodology manuscript with independently realized V2 canonical produc-
621 tion; CAPS V1 production outputs are not used as canonical evidence for
622 the present AMPV V2 results. Related CAPS V1 archive DOI: 10.5281/zen-
623 odo.22996331.

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